

Unit 2 Reference Sheet

Domain and range

Graph shows	Write	Example
closed dot	[] (included)	$-3 \leq x \leq 5 \rightarrow [-3, 5]$
open dot	() (not included)	$-3 \leq x < 5 \rightarrow [-3, 5)$
arrow	∞ always gets ()	$y \geq -4 \rightarrow [-4, \infty)$
separate points	list in braces	$\{-2, 0, 3\}$

Function notation

- $f(x)$ is read "f of x": the output when the input is x.
- Evaluate $f(3)$: put 3 in for x. Solve $f(x) = 3$: the output is 3 — find x.

Key features

Zeros (x-intercepts) • y-intercept • intervals of increase / decrease (x-values, parentheses) • relative maximum / minimum (not at an endpoint) • domain • range.

Function operations

Operation	Rule
Sum	$(f + g)(x) = f(x) + g(x)$
Difference	$(f - g)(x) = f(x) - g(x)$ — distribute the negative
Product	$(f \cdot g)(x) = f(x) \cdot g(x)$
Quotient	$(f/g)(x) = f(x)/g(x), g(x) \neq 0$

Composition

$(g \circ f)(x) = g(f(x))$, "g of f of x." Work inside first: the output of f becomes the input of g. Order matters: $g(f(x))$ and $f(g(x))$ are usually different.

Absolute value equations

1. Isolate the absolute value. 2. If it equals a negative number: no solution. 3. Otherwise set the inside equal to the positive and the negative value. 4. Solve both; write the solution set $\{..., ...\}$. Check: the solutions are where $y = |...|$ meets the horizontal line.

Parent functions and vertex form $y = \pm a \cdot f(x - h) + k$

Parent	Key point	Pattern from the key point	Domain	Range ($a > 0$)
$y = x $	vertex (h, k)	over 1, up a (both sides)	$(-\infty, \infty)$	$[k, \infty)$
$y = x^2$	vertex (h, k)	over 1 up a; over 2 up 4a (both sides)	$(-\infty, \infty)$	$[k, \infty)$
$y = \sqrt{x}$	starting point (h, k)	over 1 up a; over 4 up 2a (right only)	$[h, \infty)$	$[k, \infty)$

h: horizontal translation ($x - h$ moves right h). k: vertical translation. $|a| > 1$ vertical stretch; $0 < |a| < 1$ vertical compression; $a < 0$ reflection across the x-axis (range flips to $(-\infty, k]$).